

Trading and Hedging a Retail Electricity Portfolio

Project Report

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The full source code and generated analytical outputs are available here:

[GitHub repository](#)

Contents

1	Introduction	2
1.1	Code and Reproducibility	2
2	Data and Portfolio Construction	2
2.1	Data	2
2.2	Customer Portfolio and Load Forecast	3
3	Theory and Methodology	4
3.1	Hourly Price Forward Curve Construction	4
3.2	Futures Selection	5
3.3	Hedge Optimization	5
3.4	Day-Ahead Shaping	6
3.5	Imbalance Settlement	7
4	Empirical Results	7
4.1	Portfolio and HPFC Results	7
4.2	Hedge Results	8
4.3	Procurement Cost Comparison	9
4.4	Cost and Risk Analysis	10
4.5	Severe Price Event	11
5	Assumptions and Limitations	11
6	Conclusion and Management Recommendation	11
A	Additional Technical Results	14
A.1	KKT System for Hedge Optimization	14
A.2	Optimized Futures Positions	14
A.3	Severe Price Event	15

1 Introduction

Electricity retailers face a procurement problem in which customer demand must be supplied continuously while wholesale electricity prices can vary substantially across hours, days, and seasons. The challenge is therefore not only to purchase sufficient energy, but also to manage the price and volume risks that arise from differences between forecast demand, available hedge products, and realised customer consumption.

This report analyses a simplified retail electricity portfolio for delivery in 2024. The objective is to construct a customer load forecast, derive a market-consistent Hourly Price Forward Curve (HPFC), and evaluate alternative futures-hedging strategies. Three procurement strategies are considered: an unhedged position, a coarse annual hedge using Base and Peak futures, and a more granular hedge based on non-overlapping monthly and quarterly Base and Peak contracts.

The main difficulty is that standard electricity futures do not perfectly match the shape of the forecast customer load. This creates shape risk even when the hedge is calibrated to match the economic value of the expected portfolio. In addition, realised customer demand can differ from the Day-Ahead schedule, creating imbalance exposure that cannot be eliminated through fixed-volume futures contracts alone.

The analysis therefore separates the problem into several stages. First, the customer portfolio is converted into a quarter-hourly load forecast. Second, the HPFC is constructed from the available futures prices and historical intraday shape information. Third, the futures positions are optimized subject to value neutrality and non-negative positions. The remaining forecast-load mismatch is settled in the Day-Ahead market, while deviations between forecast and realised demand are settled using imbalance prices.

The empirical analysis compares the strategies using both cost and risk measures. Particular attention is paid to the distinction between the ex-ante hedge-design problem and the ex-post 2024 backtest. The report also examines monthly procurement-cost stability and a severe Day-Ahead price event in order to assess how the different hedge structures perform under stressed market conditions.

The analysis is guided by two main questions. First, how effectively can standard Base and Peak futures contracts replicate the shape and economic value of the forecast customer load? Second, how do different hedge structures perform when evaluated ex post using realised Day-Ahead prices, actual customer demand, and imbalance prices? These questions allow the study to distinguish between hedge quality at the design stage and realised performance during delivery.

1.1 Code and Reproducibility

The full source code and generated analytical outputs are available here: [GitHub repository](#).

The repository contains the complete notebook workflow, beginning with data preparation and portfolio construction and continuing through HPFC calibration, hedge optimization, cost accounting, and the final management analysis.

2 Data and Portfolio Construction

2.1 Data

The analysis combines several datasets covering the customer portfolio, forward-market information, realised wholesale prices, and realised electricity consumption. All time series are aligned on a common quarter-hourly UTC time index for 2024, while local timestamps in the Europe/Berlin time zone are used for market-hour classifications such as Peak and Off-Peak periods.

The customer-side input consists of standardized load profiles for three customer groups: households (HB), commercial customers (GB), and agricultural customers (LB). These profiles

are used to determine the intraday and seasonal shape of electricity demand. The market-side inputs include futures prices observed on 29 September 2023, historical shape factors used for the Hourly Price Forward Curve construction, realised 2024 Day-Ahead prices, realised portfolio load, and imbalance prices.

Before the empirical analysis, the datasets are checked for completeness of timestamps, alignment across sources, and consistency of the local-time conversion. The analysis uses 35,136 quarter-hourly observations for the full leap year 2024. The hourly Day-Ahead price series is expanded to quarter-hourly frequency so that it can be matched consistently with the load and hedge time series.

The standardized load profiles for households, commercial customers, and agricultural customers are provided by Stromnetz Berlin GmbH. Day-Ahead prices and the historical price-shape inputs are sourced from the Bundesnetzagentur SMARD platform, while imbalance prices (reBAP) are obtained from Netztransparenz and the German transmission system operators. The futures snapshot and the realised portfolio-load dataset are synthetic teaching inputs provided as part of the HTW Summer School project.

Table 1: Data sources used in the empirical analysis

Dataset	Source
Standard load profiles (HB, GB, LB)	Stromnetz Berlin GmbH
Day-Ahead prices and shape inputs	Bundesnetzagentur / SMARD.de
Imbalance prices (reBAP)	Netztransparenz / German TSOs
Futures snapshot	HTW Summer School project (synthetic)
Actual portfolio load	HTW Summer School project (synthetic)

2.2 Customer Portfolio and Load Forecast

The retail portfolio consists of three customer groups with different annual consumption levels and load shapes. The household segment is the dominant component of the portfolio, followed by commercial and agricultural customers. The annual portfolio demand is constructed from the number of customers and the assumed annual energy consumption per customer.

The household segment contains 10,000 customers with an annual consumption of 3.5 MWh per customer, resulting in 35,000 MWh. The commercial segment contains 200 customers with 30 MWh per customer, resulting in 6,000 MWh. The agricultural segment contains 20 customers with 100 MWh per customer, resulting in 2,000 MWh. Total forecast annual consumption is therefore 43,000 MWh.

Table 2: Customer portfolio composition

Customer Group	Customers	MWh/Customer	Annual MWh	Share
Households (HB)	10,000	3.5	35,000	81.40%
Commercial (GB)	200	30.0	6,000	13.95%
Agricultural (LB)	20	100.0	2,000	4.65%
Total	10,220	–	43,000	100.00%

The standardized load profiles are scaled so that the annual energy of each customer group exactly matches its corresponding portfolio target. The resulting quarter-hourly forecast is obtained by summing the three scaled group profiles. This produces an internally consistent demand forecast while preserving the characteristic temporal shape of each customer segment.

As shown in Table 2, the portfolio is strongly household-dominated: households account for approximately 81.4% of forecast annual demand, commercial customers for 14.0%, and agri-

cultural customers for 4.7%. Consequently, the aggregate load profile displays pronounced residential morning and evening demand patterns, while the commercial component contributes relatively more during daytime hours. These differences in intraday shape are important for the hedging problem because standard Base and Peak futures cannot replicate the quarter-hourly customer load exactly.

Figure 1 shows the distinct intraday demand patterns of the three customer groups and the resulting aggregate portfolio shape.

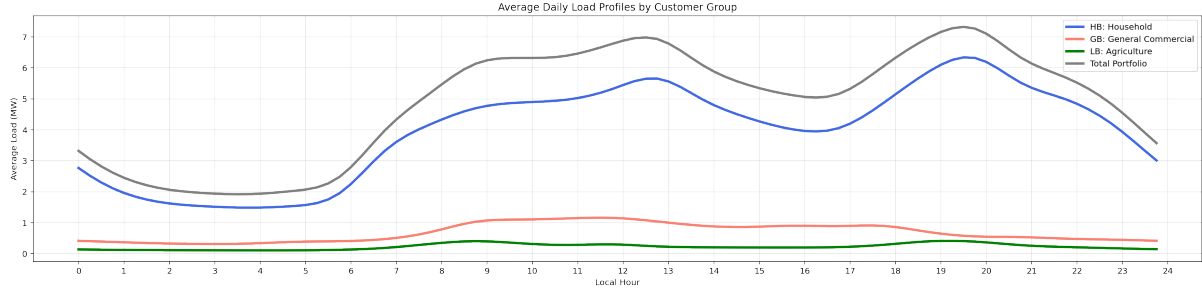


Figure 1: Average daily load profiles by customer group

As a final validation step, the constructed load forecast is reconciled against the target annual energy of each customer group. The household, commercial, and agricultural profiles sum to 35,000 MWh, 6,000 MWh, and 2,000 MWh respectively, yielding the intended total of 43,000 MWh.

3 Theory and Methodology

3.1 Hourly Price Forward Curve Construction

The available futures market provides prices only for a limited set of delivery periods and load types, while the subsequent portfolio valuation and hedge optimization require prices at a much finer time resolution. Therefore, an Hourly Price Forward Curve (HPFC) is constructed to translate the observed futures prices into a quarter-hourly valuation curve.

This approach follows the general logic of Fleten and Lemming [2003], who show that electricity forward curves can be constructed by combining information from traded market products with additional information about the expected seasonal or temporal price shape. In electricity markets, the limited number of quoted forward and futures contracts does not directly reveal a complete high-resolution term structure, which motivates the use of supplementary shape information.

In this project, historical shape factors provide the relative intraday and seasonal pattern, while the observed futures prices determine the absolute price level. For each delivery block, separate Peak and Off-Peak price levels are derived. Peak intervals are defined as Monday to Friday from 08:00 to 20:00 local time, while all remaining intervals are treated as Off-Peak.

The implied Off-Peak price is obtained from the relationship between the Base and Peak contracts. Let P^B denote the Base price, P^P the Peak price, and N_{all} , N_{peak} , and N_{off} the number of intervals in the respective delivery block. The implied Off-Peak price is

$$P^O = \frac{P^B N_{\text{all}} - P^P N_{\text{peak}}}{N_{\text{off}}}. \quad (1)$$

The historical shape factors are then scaled separately within the Peak and Off-Peak segments so that their average prices reproduce the relevant market-implied price levels. Consequently, the resulting HPFC preserves the detailed temporal shape while remaining consistent with the futures contracts used for calibration.

Importantly, the HPFC is used as a *forward valuation curve* rather than as a forecast of realised 2024 Day-Ahead prices. Its purpose is to provide a market-consistent price signal for valuing the forecast customer load and for constructing the futures hedge.

3.2 Futures Selection

The futures are filtered to retain contracts with sufficient observed market activity. This reduces the risk of basing the hedge on illiquid observations and keeps the strategy aligned with products that would plausibly be available for implementation.

Two hedge structures are considered. The *COARSE_CAL* strategy uses a single annual Base contract together with an annual Peak contract, providing a simple year-wide hedge with only two positions.

The *GRANULAR* strategy uses the finest non-overlapping set of eligible delivery periods. Monthly contracts M01, M02, and M03 are used for the first quarter, while Q2, Q3, and Q4 cover the remainder of the year. Base and Peak positions are determined separately within each block, allowing the hedge level to vary across the year while avoiding overlapping positions.

The distinction between the two strategies is important for the subsequent optimization. *COARSE_CAL* can only adjust exposure between Base and Peak hours over the full year, whereas *GRANULAR* can additionally vary the hedge across months and quarters. This greater temporal flexibility is expected to reduce the mismatch between the futures hedge and the forecast customer load, although it does not guarantee lower realised procurement cost.

Figure 2 summarises market activity across the available futures delivery periods for both Base and Peak contracts. A threshold of 100 observed trades is used as a simple liquidity screen. Annual, quarterly, and the first three monthly products exceed this threshold, while later monthly contracts show substantially lower activity. This motivates the use of *CAL* for the coarse hedge and the non-overlapping set M01, M02, M03, Q2, Q3, and Q4 for the granular hedge.

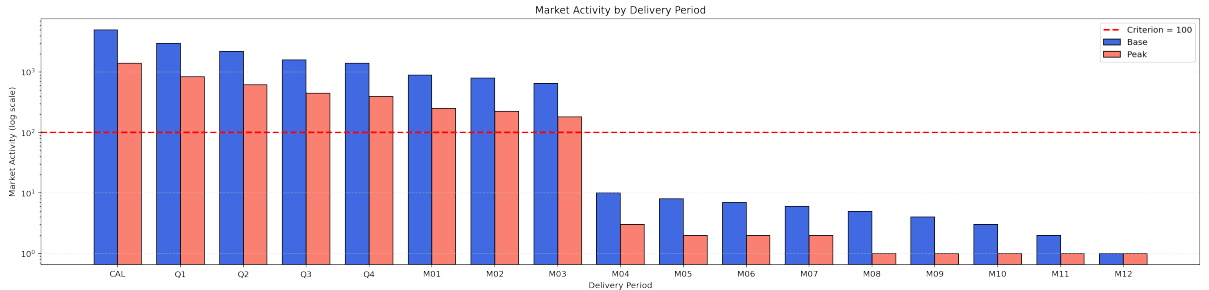


Figure 2: Market activity by delivery period for Base and Peak futures

3.3 Hedge Optimization

For each hedge strategy, the futures positions are chosen to approximate the quarter-hourly forecast load as closely as possible while preserving the economic value of the forecast portfolio under the HPFC.

Let L_t denote forecast electricity demand in interval t , and let H_t denote the energy delivered by the futures hedge. For delivery block b , the optimization minimizes the squared difference between forecast load and hedge delivery,

$$\min_{\mathbf{x}_b} \frac{1}{2} (\mathbf{L}_b - D_b \mathbf{x}_b)^T (\mathbf{L}_b - D_b \mathbf{x}_b), \quad (2)$$

where $\mathbf{x}_b$ contains the Base and Peak futures positions and D_b is the corresponding delivery matrix.

The Base position contributes to every quarter-hour interval in the delivery block, while the Peak position is incremental and contributes only during Peak hours. Since each interval represents 15 minutes, hedge energy is

$$H_t = 0.25 \left(x_{B,b} + I_t^P x_{P,b} \right), \quad (3)$$

where $I_t^P = 1$ during Peak intervals and 0 otherwise.

The optimization is subject to a value-neutrality constraint. The HPFC-weighted value of the hedge must equal the HPFC-weighted value of the forecast load,

$$\sum_t H_t \pi_t = \sum_t L_t \pi_t, \quad (4)$$

where π_t denotes the HPFC price in interval t .

Futures positions are additionally constrained to be non-negative,

$$x_{B,b} \geq 0, \quad x_{P,b} \geq 0. \quad (5)$$

The resulting constrained quadratic optimization problem is solved using the Karush–Kuhn–Tucker (KKT) conditions. The implementation evaluates the interior solution as well as the Base-only and Peak-only boundaries and retains the feasible solution with the smallest squared forecast-load mismatch.

The same optimization logic is applied to the annual CAL block for the COARSE_CAL strategy and separately to $M01$, $M02$, $M03$, $Q2$, $Q3$, and $Q4$ for the GRANULAR strategy. The full KKT system is reported in the Appendix.

3.4 Day-Ahead Shaping

After the futures hedge has been determined, the remaining difference between the forecast customer load and the hedge delivery must be managed in the Day-Ahead market. Let L_t denote forecast load and H_t the hedge energy in interval t . The residual Day-Ahead position is

$$R_t^{DA} = L_t - H_t. \quad (6)$$

A positive residual indicates that additional energy must be purchased in the Day-Ahead market, while a negative residual indicates that the hedge exceeds the forecast load and the surplus is treated as a Day-Ahead sale.

The forecast procurement cost can therefore be written as

$$C_{\text{forecast}} = C_{\text{futures}} + \sum_t R_t^{DA} P_t^{DA}, \quad (7)$$

where P_t^{DA} denotes the realised Day-Ahead price and C_{futures} is the cost of the futures positions.

An equivalent financial representation is

$$C_{\text{forecast}} = \sum_t L_t P_t^{DA} - \Pi_{\text{futures}}, \quad (8)$$

where the futures payoff is defined as

$$\Pi_{\text{futures}} = \sum_t H_t P_t^{DA} - C_{\text{futures}}. \quad (9)$$

Both representations are algebraically equivalent and are used as a consistency check in the implementation.

The Day-Ahead shaping stage therefore captures the residual price exposure that remains after the futures hedge. A hedge that more closely follows the forecast load reduces the absolute volume that must be settled at Day-Ahead prices, although this does not necessarily imply a lower realised total procurement cost.

3.5 Imbalance Settlement

The Day-Ahead schedule is based on the forecast customer load, while realised consumption may differ from this forecast. Let A_t denote the actual portfolio load and L_t the Day-Ahead schedule based on the forecast. The resulting imbalance volume is

$$I_t = A_t - L_t. \quad (10)$$

A positive imbalance indicates that actual consumption exceeds the scheduled volume, whereas a negative imbalance indicates that the portfolio consumed less than forecast.

These deviations are settled at the realised imbalance price P_t^I . The total imbalance settlement is therefore

$$C_{\text{imbalance}} = \sum_t I_t P_t^I. \quad (11)$$

To isolate the economic effect of imbalance pricing relative to the Day-Ahead market, the imbalance premium is defined as

$$\Pi_{\text{imbalance}} = \sum_t I_t (P_t^I - P_t^{DA}). \quad (12)$$

The final realised procurement cost is then obtained by adding the imbalance settlement to the forecast procurement cost,

$$C_{\text{final}} = C_{\text{forecast}} + C_{\text{imbalance}}. \quad (13)$$

An important feature of this setup is that the imbalance volume is identical across the three hedge strategies. Futures positions change the financial exposure to wholesale prices, but they do not change the physical Day-Ahead schedule, which remains equal to the forecast load.

This remaining quantity risk is consistent with the discussion of Oum et al. [2006], who show that fixed-volume forward hedges cannot fully eliminate the exposure faced by a load-serving entity when electricity demand is uncertain. The uncertainty in realised consumption therefore remains an important source of risk even after the forecast portfolio has been hedged with futures.

4 Empirical Results

4.1 Portfolio and HPFC Results

The first empirical step evaluates whether the constructed HPFC is consistent with the futures contracts used for calibration. The GRANULAR specification is calibrated separately to the selected monthly and quarterly delivery blocks, with Base and Peak prices imposed as market anchors.

Table 3: HPFC calibration against selected GRANULAR futures contracts

Period	Base Quote	HPFC Base	Peak Quote	HPFC Peak
M01	75.32	75.32	88.68	88.68
M02	60.09	60.09	70.59	70.59
M03	63.45	63.45	72.79	72.79
Q2	66.23	66.23	64.76	64.76
Q3	74.74	74.74	69.07	69.07
Q4	101.39	101.39	134.42	134.42

As shown in Table 3, the granular HPFC reproduces all selected Base and Peak futures quotes within numerical precision. This confirms that the quarter-hourly valuation curve remains consistent with the market contracts used for calibration.

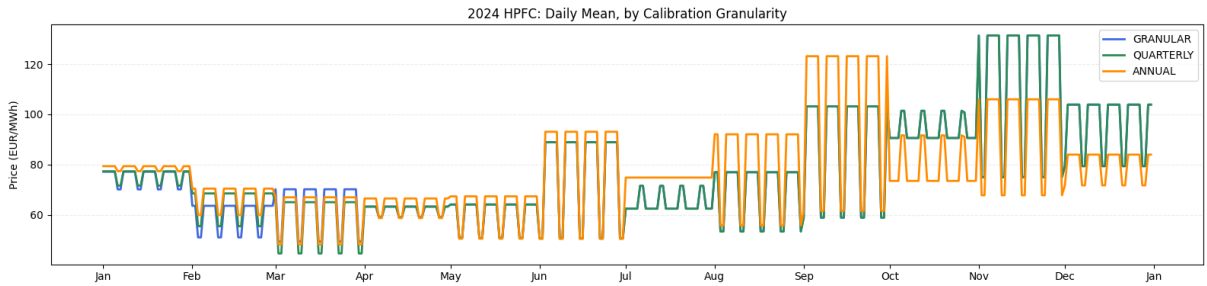


Figure 3: Daily mean HPFC by calibration granularity

Figure 3 shows that the calibration granularity affects the temporal structure of the forward curve. The annual specification imposes the coarsest price structure, while the quarterly and granular specifications allow greater variation across delivery periods. The GRANULAR curve therefore retains more detailed information from the selected monthly and quarterly futures contracts used in the hedge design.

4.2 Hedge Results

The hedge optimization produces non-negative Base and Peak positions for both futures-based strategies while satisfying the HPFC value-neutrality constraint within numerical precision. The GRANULAR strategy achieves a closer match to the forecast customer load than the COARSE_CAL strategy.

The complete set of optimized Base and Peak futures positions is reported in Table 7 in the Appendix.

Table 4 compares the main hedge-quality measures. Although the two strategies have similar total hedge notionals, the GRANULAR hedge produces both a lower residual-load RMSE and a lower absolute residual Day-Ahead volume. This is consistent with the additional temporal flexibility provided by the monthly and quarterly delivery blocks.

Table 4: Hedge-quality comparison

Strategy	Hedge Notional (MWh)	Residual RMSE (MWh)	Absolute Residual DA Volume (MWh)
UNHEDGED	0.00	1.3192	43,000.00
COARSE_CAL	44,503.82	0.4417	12,760.13
GRANULAR	44,079.18	0.4213	11,985.15

Relative to COARSE_CAL, the GRANULAR strategy reduces the residual-load RMSE by approximately 4.6% and the absolute residual Day-Ahead volume by approximately 6.1%. This confirms that the more granular delivery structure improves the ex-ante shape match between the futures hedge and the forecast portfolio.

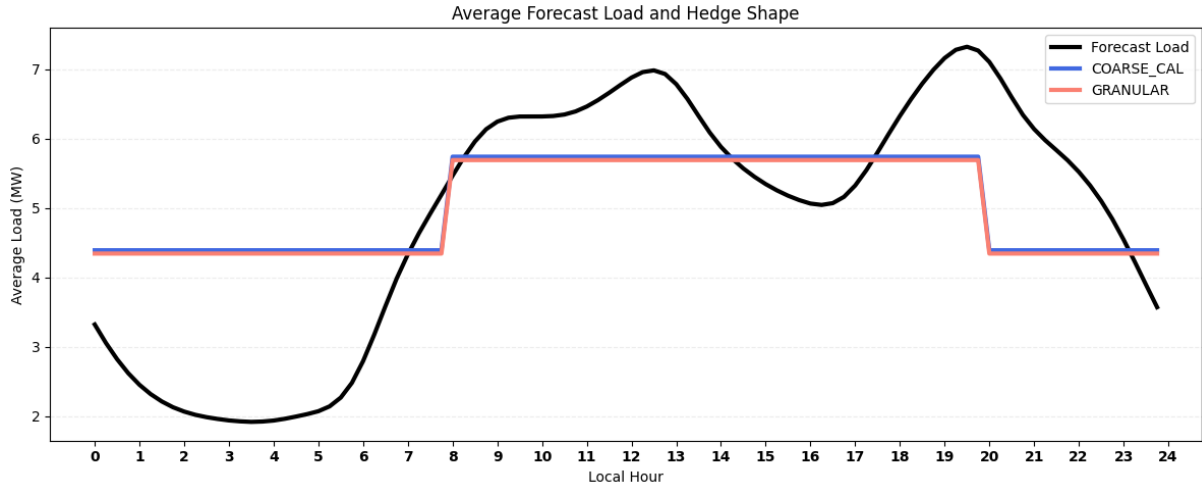


Figure 4: Average forecast load and hedge shape for the COARSE_CAL and GRANULAR

Figure 4 shows that both futures-based strategies reproduce the broad Base/Peak structure of the forecast portfolio, but neither can match the continuous intraday load shape exactly. The differences between COARSE_CAL and GRANULAR appear relatively small in the annual average daily profile because seasonal adjustments are averaged across the year. Their difference is captured more clearly by the residual-load metrics reported in Table 4.

4.3 Procurement Cost Comparison

The realised 2024 procurement costs are evaluated after combining futures costs, residual Day-Ahead transactions, and imbalance settlement. Since all three strategies use the same forecast as the Day-Ahead schedule, the imbalance settlement is identical across strategies; the cost differences therefore arise from the futures hedge and the resulting Day-Ahead residual exposure.

Table 5 summarises the main realised cost outcomes.

Table 5: Realised 2024 procurement cost comparison

Strategy	Forecast Cost (EUR)	Imbalance (EUR)	Final Cost (EUR)	Avg. Price (EUR/MWh)
UNHEDGED	3,506,703.99	30,503.84	3,537,207.83	82.00
COARSE_CAL	3,450,966.64	30,503.84	3,481,470.48	80.71
GRANULAR	3,451,507.34	30,503.84	3,482,011.18	80.72

Both futures-based strategies reduce realised annual procurement cost relative to the UNHEDGED case. COARSE_CAL achieves a saving of approximately EUR 55,737, while GRANULAR achieves a saving of approximately EUR 55,197. The difference between the two hedged strategies is therefore small in annual cost terms, with COARSE_CAL producing the slightly lower realised cost.

The identical imbalance settlement of approximately EUR 30,504 across all strategies reflects the fact that futures positions do not alter the physical Day-Ahead schedule or the realised forecast error. This result reinforces the distinction between price hedging and quantity risk:

the futures hedge affects the financial exposure to wholesale prices, but the deviation between actual and forecast demand remains unchanged.

Although GRANULAR provides the better ex-ante shape match, the realised 2024 price path results in a marginally lower annual procurement cost for COARSE_CAL.

4.4 Cost and Risk Analysis

Annual procurement cost alone does not fully describe the risk characteristics of the three strategies. To assess cost stability, the analysis also compares the monthly average procurement price across 2024.

Table 6 reports the standard deviation and range of monthly procurement prices for each strategy.

Table 6: Monthly procurement-price risk measures		
Strategy	Monthly Std. Dev. (EUR/MWh)	Monthly Range (EUR/MWh)
UNHEDGED	20.15	58.43
COARSE_CAL	2.43	9.08
GRANULAR	18.56	47.08

The results show a pronounced difference in realised cost stability. COARSE_CAL produces a monthly procurement-price standard deviation of only 2.43 EUR/MWh, compared with 20.15 EUR/MWh for the UNHEDGED strategy and 18.56 EUR/MWh for GRANULAR. The same pattern is visible in the monthly price range, which is substantially narrower for COARSE_CAL.

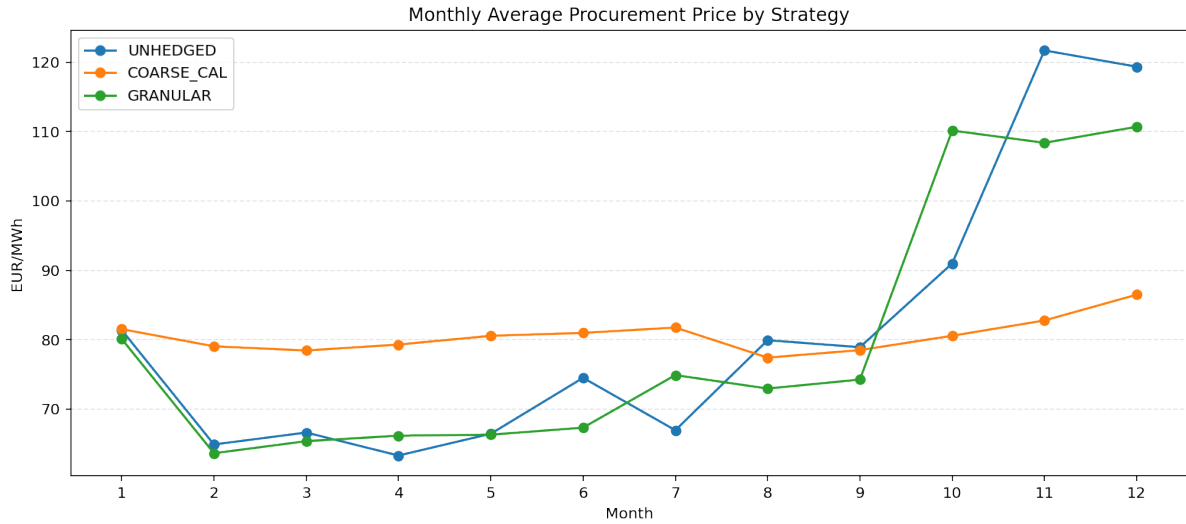


Figure 5: Monthly average procurement price by strategy

Figure 5 confirms that the annual hedge substantially smooths realised monthly procurement prices. By contrast, the GRANULAR strategy remains more exposed to month-to-month price variation despite its better ex-ante shape match.

This illustrates the importance of distinguishing between hedge quality at the design stage and realised performance during the delivery year. Ex ante, GRANULAR provides the closer match to forecast load and therefore reduces residual Day-Ahead exposure. Ex post, however, the realised 2024 price path favours COARSE_CAL, which achieves both a slightly lower annual cost and substantially greater monthly cost stability.

The result should therefore be interpreted as a realised backtest outcome rather than evidence that the annual strategy was known in advance to be superior.

4.5 Severe Price Event

To examine the behaviour of the hedge strategies under stressed market conditions, the analysis considers the most severe realised Day-Ahead price event in 2024. The event is selected mechanically as the interval with the maximum observed Day-Ahead price, rather than chosen subjectively after inspecting the hedge results.

The maximum Day-Ahead price occurs on 12 December 2024 at 17:00 local time and reaches 936.28 EUR/MWh. At such an extreme price, the amount of residual load remaining exposed to the Day-Ahead market becomes particularly important. The UNHEDGED strategy leaves the full forecast load exposed, whereas the futures-based strategies reduce this exposure to the residual difference between forecast load and hedge delivery.

A detailed illustration of the event, including the realised Day-Ahead price, hedge profiles, and residual exposure, is provided in Figure 6 in the Appendix.

The event shows that futures hedging can materially reduce exposure to extreme spot prices, but it does not eliminate risk completely. Standard Base and Peak contracts cannot reproduce the quarter-hourly customer load exactly, while deviations between actual and forecast consumption remain subject to imbalance settlement.

5 Assumptions and Limitations

The analysis relies on several simplifying assumptions. First, the customer load forecast is constructed from standardized load profiles scaled to fixed annual consumption targets. This preserves realistic intraday and seasonal shapes, but does not model customer-specific behavioural changes or forecast uncertainty explicitly.

Second, the HPFC is derived from a limited set of futures contracts and historical shape factors. Although the calibration reproduces the selected market prices exactly, the resulting quarter-hourly curve is still a model-based allocation of forward value and should not be interpreted as a forecast of realised Day-Ahead prices.

Third, the hedge is restricted to non-negative Base and Peak futures positions. This reflects a procurement-oriented strategy and avoids speculative short positions, but it also limits the set of feasible hedge shapes.

Fourth, the optimization is based on the forecast portfolio and ex-ante market information available on 29 September 2023. Realised 2024 Day-Ahead prices, imbalance prices, and actual load enter only in the backtest. Consequently, the superior realised stability of COARSE_CAL cannot be interpreted as information that would have been available when the hedge was designed.

Finally, the analysis does not include transaction costs, bid-ask spreads, collateral requirements, trading constraints beyond the activity filter, or counterparty and liquidity risk. Including these features could materially affect the relative attractiveness of the strategies in practice.

6 Conclusion and Management Recommendation

This report develops and evaluates a simplified electricity procurement framework for a retail portfolio with 43,000 MWh of forecast annual demand. A quarter-hourly load forecast is combined with a market-consistent HPFC, and two futures-based hedge strategies are compared with an unhedged benchmark.

The empirical results show that both futures strategies substantially reduce residual Day-Ahead exposure and realised procurement cost relative to the unhedged case. From an ex-

ante hedge-design perspective, GRANULAR performs better: its monthly and quarterly delivery blocks reduce residual-load RMSE and absolute residual Day-Ahead volume relative to COARSE_CAL.

The realised 2024 backtest, however, produces a different ranking in terms of cost stability. COARSE_CAL achieves a slightly lower annual procurement cost and a much lower standard deviation and range of monthly procurement prices. This illustrates that a closer physical hedge match does not necessarily translate into lower realised cost under a particular future price path.

For management, the two strategies should therefore be interpreted according to their objective. GRANULAR is preferable when the priority is to match the expected customer load more closely and reduce residual Day-Ahead volume. COARSE_CAL, by contrast, delivered greater realised cost stability in the 2024 backtest and would therefore have been the more stable strategy under the observed market outcome.

The results also confirm that futures hedging does not eliminate all risk. Forecast errors remain exposed to imbalance settlement, while standard Base and Peak contracts cannot reproduce the full quarter-hourly customer load shape. Future extensions could incorporate stochastic load scenarios, transaction costs, liquidity constraints, alternative derivatives, or explicit risk measures such as Value-at-Risk or Expected Shortfall.

References

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- Yumi Oum, Shmuel Oren, and Shijie Deng. Hedging quantity risks with standard power options in a competitive wholesale electricity market. *Naval Research Logistics*, 53(7):697–712, 2006.

A Additional Technical Results

A.1 KKT System for Hedge Optimization

For an interior solution, the first-order conditions are

$$\begin{bmatrix} D_b^T D_b & a_b \\ a_b^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}_b \\ \lambda_b \end{bmatrix} = \begin{bmatrix} D_b^T \mathbf{L}_b \\ c_b \end{bmatrix}, \quad (14)$$

with

$$a_b = D_b^T \boldsymbol{\pi}_b, \quad c_b = \mathbf{L}_b^T \boldsymbol{\pi}_b.$$

A.2 Optimized Futures Positions

Table 7 reports the optimized Base and Peak futures positions used in the two hedge strategies. Peak positions are incremental to the corresponding Base positions.

Table 7: Optimized futures positions

Strategy	Period	Load Type	Position (MW)	Notional (MWh)	Price (EUR/MWh)
COARSE_CAL	CAL	BASE	4.3920	38,579.12	77.26
COARSE_CAL	CAL	PEAK	1.8844	5,924.70	86.60
GRANULAR	M01	BASE	5.0437	3,752.48	75.32
GRANULAR	M01	PEAK	2.3310	643.35	88.68
GRANULAR	M02	BASE	4.9331	3,433.43	60.09
GRANULAR	M02	PEAK	2.2172	558.74	70.59
GRANULAR	M03	BASE	4.6509	3,455.61	63.45
GRANULAR	M03	PEAK	1.9714	496.79	72.79
GRANULAR	Q2	BASE	4.0000	8,735.97	66.23
GRANULAR	Q2	PEAK	1.6782	1,309.02	64.76
GRANULAR	Q3	BASE	3.8483	8,496.94	74.74
GRANULAR	Q3	PEAK	1.5412	1,220.64	69.07
GRANULAR	Q4	BASE	4.6561	10,285.43	101.39
GRANULAR	Q4	PEAK	2.1348	1,690.78	134.42

A.3 Severe Price Event

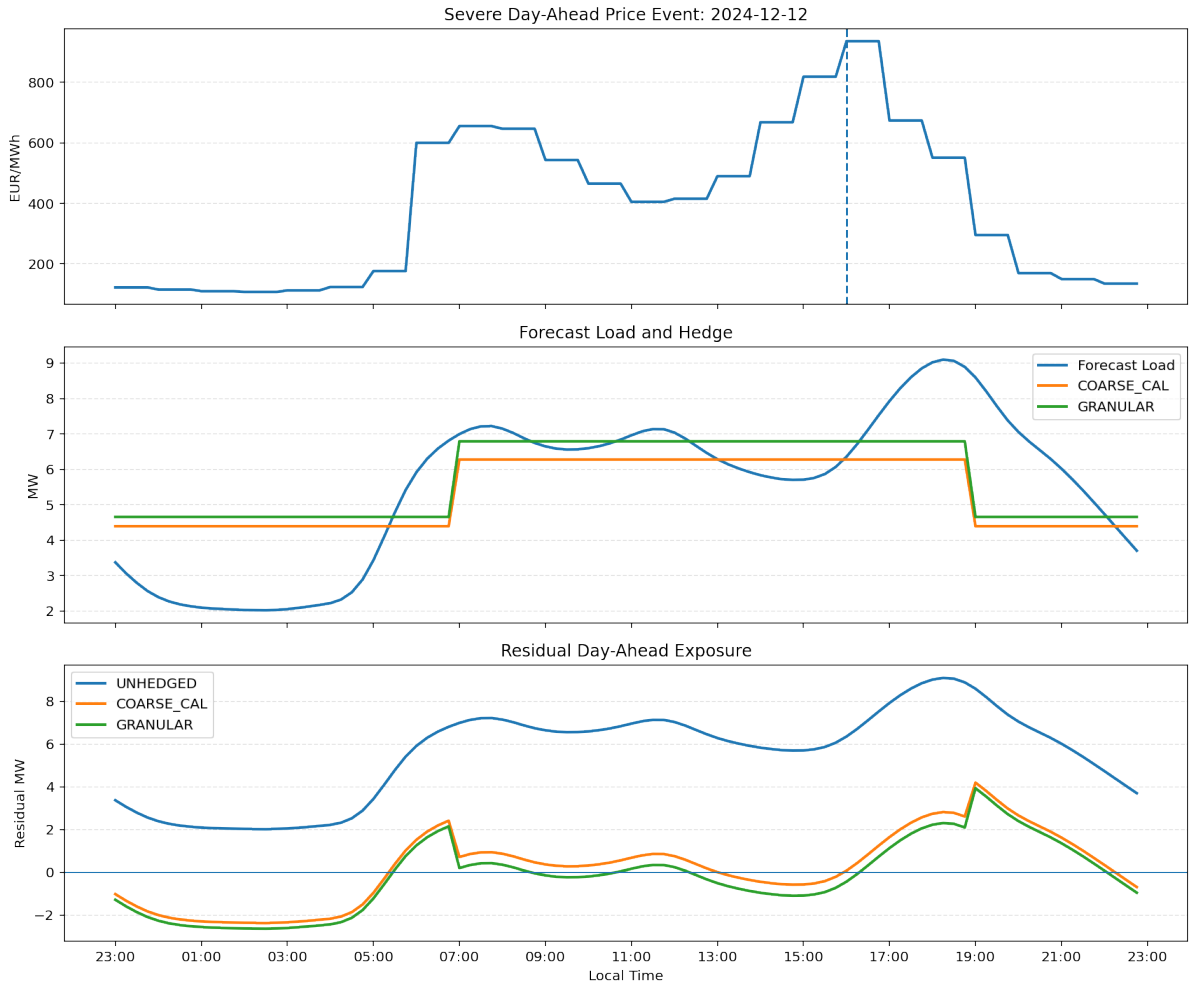


Figure 6: Hedge behaviour during the severe Day-Ahead price event on 12 December 2024